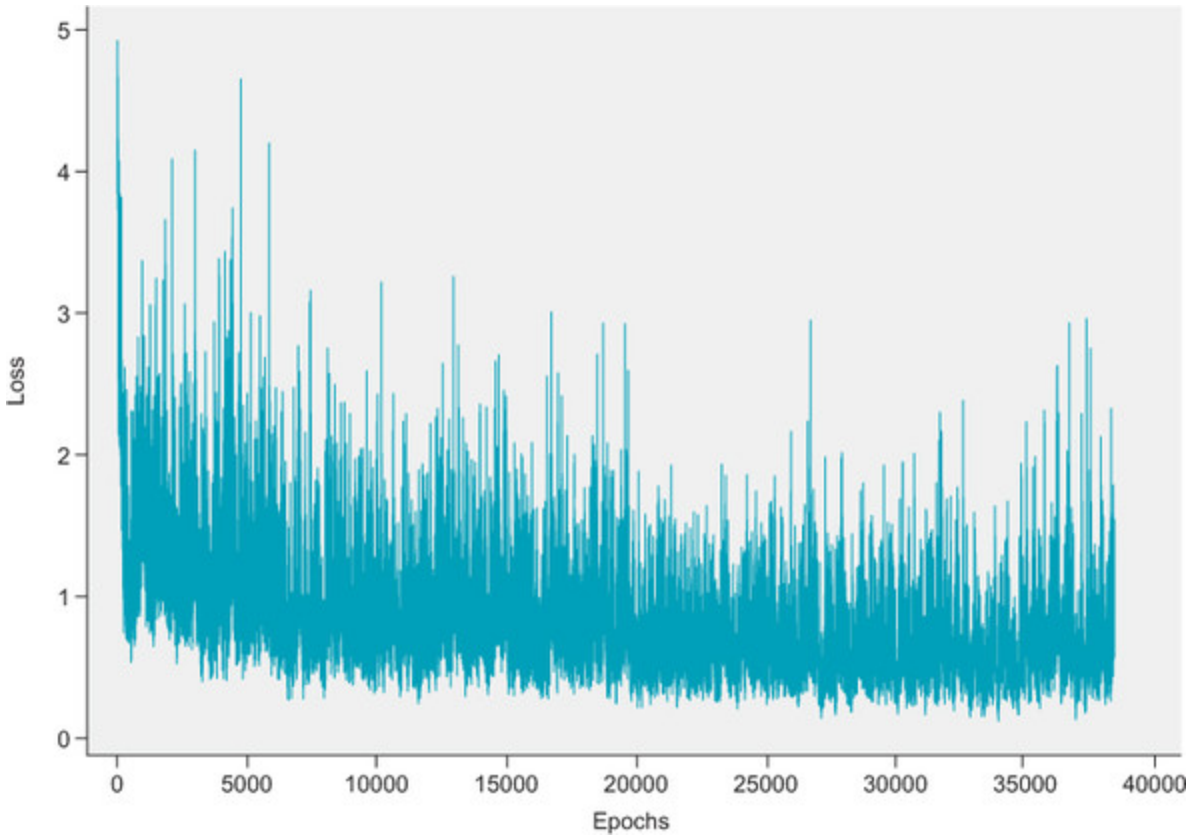


In the loss in figure 3.14, you can see it's definitely trending downward, but it looks pretty unstable. This is the type of plot you'd be a bit surprised to see in a supervised learning problem, but it's quite common in bare DRL. The experience replay mechanism helps with training stabilization by reducing catastrophic forgetting, but there are other related sources of instability.

3.4. Improving stability with a target network

So far, we've been able to successfully train a deep reinforcement learning algorithm to learn and play Gridworld with both a deterministic static initialization and a slightly harder version where the player is placed randomly on the board each game. Unfortunately, even though the algorithm appears to learn how to play, it is quite possible it is just memorizing all the possible board configurations, since there aren't that many on a 4 × 4 board. The hardest variant of the game is where the player, goal, pit, and wall are all initialized randomly each game, making it much